

InFringement

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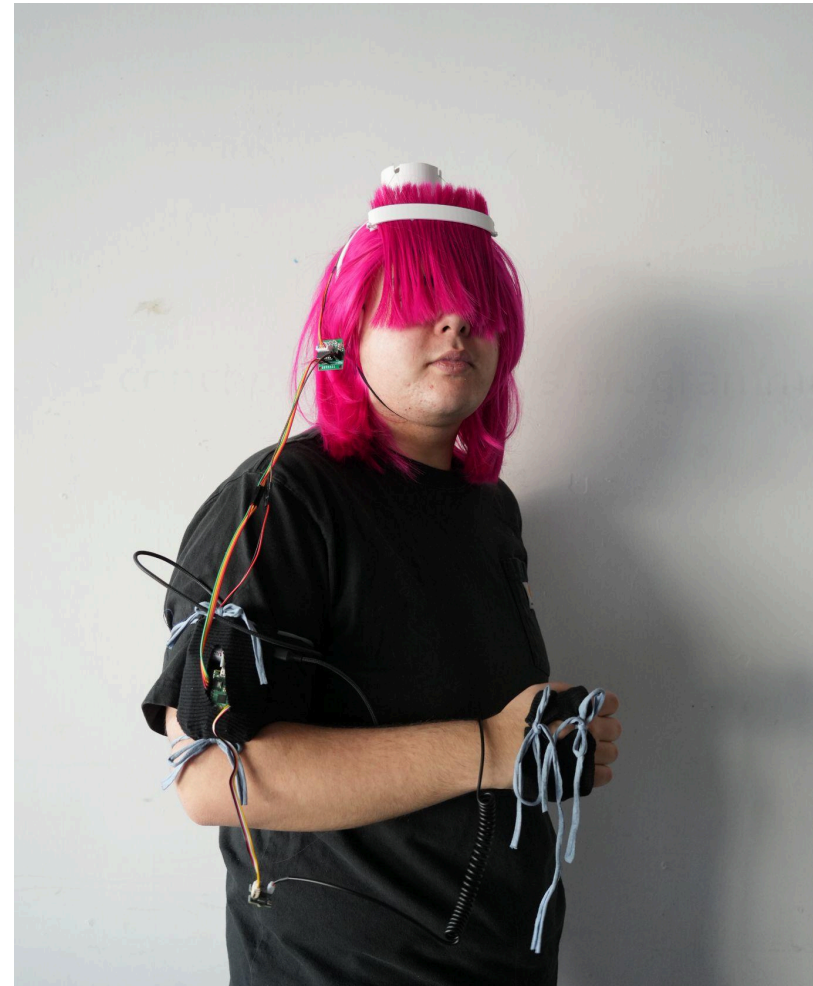
Elio Bidinost CART 360

<https://github.com/mvlamis/wig>

Abstract

A wig can be anything a wig wants to be. It can be playful, theatrical, and transformative, but it can also make us more cognizant of how we are being observed. *InFringement* is an interactive wearable artifact that explores the relationship between gender performativity, self-awareness, and social interaction. The project manifests itself in the form of a bright pink wig equipped with biofeedback sensors and a mechanical bang system that reacts to the participant's internal state. Using galvanic skin response (GSR) sensors and electroencephalogram (EEG) sensors, the created system translates typically subtle physiological changes into visible movement, raising and lowering the bangs to alternately conceal and reveal the wearer's eyes.

The artifact is designed as a shared experience between two volunteers, who observe each other and interact while wearing the reactive wigs. By externalizing internal states, the project makes normally imperceptible emotional responses visible in real time, pushing the participants to reflect on how they present themselves and respond to social pressure.



Through a non-linear, iterative research-creation process that involved conceptual development, technical prototyping, and physical construction, *InFringement* evolved from a reactive object into an embodied, interactive experience. The final artifact finds balance between absurdity and vulnerability through the use of exaggerated feminine aesthetics and responsive mechanics to create a setting where participants can engage playfully while confronting the structured nature of gender expression and identity.

Design Narrative



Rear



Front



Side



There is more to a wig than meets the eye. One may wear a wig to perform an identity, embody a character, or simply try something new. *InFringement* began from our interest in the relationship between gender presentation and self-consciousness. We wanted to make an artifact that could make invisible feelings visible while also asking participants to confront their own perception and judgment of gender in public. Our final piece takes the form of a bright pink wig fitted with a Muse headset, a GSR sensor, and a mechanical bang system that rises and falls in response to the wearer's detected anxiety or discomfort.

The conceptual foundation is rooted in Judith Butler's theory of gender performativity, the argument that gender is not an essential, innate truth that we express outwardly, but a set of norms that produce the effect of a gendered self. The choice of the wig was central from the beginning. The wig serves as an immediately legible symbol of constructed identity, in line with Butler's notion that gender is performed through repeated acts rather than expressed as an internal truth. By putting on the wig, the participants alter their appearance and engage in a visible act of gender performance. At the same time, the wig is

intentionally exaggerated and absurd. The bright artificial pink colour and absurd appearance lean into a feeling of playfulness and theatrics. There is an obvious balance between sincere vulnerability and complete ridiculousness, which is very important to this artifact because it makes participation feel more accessible, while maintaining the visibility of the deeper themes of gender performance and experimentation, ensuring that they remain present and perceptible.

The artifact also draws from femininity and drag aesthetics, where presentation is stylized and heightened, often in an exaggerated, artificial way. There is obviously no attempt at replicating “real” hair; in this way, it emphasizes the labour and intentionality behind gendered appearance, specifically femininity. By exaggerating the qualities expected in feminine appearances, this project unveils how femininity is often performed in ways that are meant to be seen and evaluated by others. This is particularly significant in the context of the experience, where participants are being observed and made aware of their own presentation in real time.

The mechanical shifting bangs further reinforce this dynamic of visibility and concealment. As the bangs move up and down in response to physiological stress signals, they cover and reveal the participants' eyes, which are a key point of connection and vulnerability in social interactions. The motion transforms typically subtle internal states into visible change, creating a constant shift between hiding and being seen by the other. The wig does come across as playful and whimsical at first, but it transforms into something that heightens the emotional intensity of the interaction between strangers. From this perspective, the artifact remains in tension between lighthearted but uncomfortable, artificial but revealing, using exaggeration to make internal experiences more socially present and visible.

Prototyping

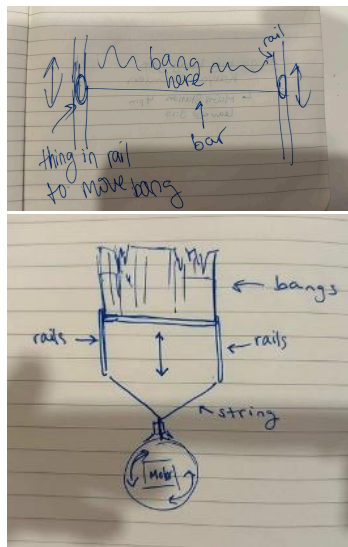
Early Concepts

At the very start of our brainstorming process, the project was not thought of as a fully formed interactive experience. Early brainstorming focused on the idea of a “reactive” wig that would move in response to specific input. At this stage, the focus was primarily on the object itself and its potential for expression.

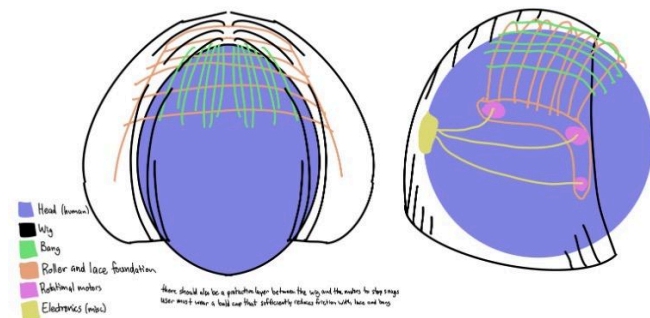
Initial concept ideas explored a variety of possible inputs that could influence the movement of the bangs. We considered environmental data like sound levels or proximity, thinking about the symbolism of the idea that “the louder you are, the more you are seen”, or inversely, that toning yourself down could result in a sort of disappearance. These early concepts were connected to questions around visibility, gender, and who is allowed to take up space in society.

As we developed the concept, we started to consider more embodied forms of input, including touch, movement, and physiological sensing. This shift introduced the possibility of using the body itself as a source of data, allowing internal states to directly influence the height of the bangs. Simultaneously, the idea expanded from a single wearable object to something leaning more towards an immersive experience between two strangers. While the object itself did not change drastically, the project went from an isolated reactive object to an interactive happening to be felt between two participants, where biofeedback is transformed into something that can be experienced on a personal and social level.

Interaction & System Development



With the initial concept properly mapped out, we moved our focus to figuring out how the system itself would actually function. One aspect that felt very important to tackle, but also very daunting, was the mechanisms that would allow the bangs to move up and down the visage of the participant. We considered using a system of conveyors that would be secured to the lace foundation of the wig. In this scenario, the bangs would be situated beneath a layer of hair, and the mechanisms involved in controlling them would be hidden. This version would have had a more seamless and subtle appearance. Another version that was mulled over was using a pair of rails that a bar would travel up and down on. In this scenario, the bangs would be attached to the bar to create the motion.



Further in the creation process, we decided to go with a simple rotating spool attached to the motor that would reel in and release string connected to both ends of a bar holding the bangs, allowing it to move up and down. At this phase, we also thought about using narrow, clear vinyl tubing to guide the string out of the case holding the motor, which would be positioned at the back of the head, and route it towards the front where the bangs would be located. This would allow the string to travel cleanly across the head while keeping the mechanism relatively contained, although at this stage, we had not yet determined how the casing itself would be secured. We were also still considering whether the tubing and string would run over or underneath the rest of the wig hair, depending on how visible we wanted the system to be.

With a clearer idea of how the system should function mechanically, we moved into testing the technical components that would drive this movement. This meant focusing on the relationship between sensor input and motor output, and experimenting with how physiological data could be translated into responsive, meaningful motion.

Technical Prototyping

The technical development of *InFringement* was incremental, beginning with the sensors before the mechanical output system. This allowed each layer of the system to be understood before having to cooperate with the next one. The GSR sensors were relatively cooperative. The sensor outputs an analog voltage that rises with increased skin conductivity, making it straightforward to map its readings onto a normalized scale representing bang height as a percentage. With a 20-second-long calibration baseline established, the sensor data could be translated directly into a value for a position. To make sense of the sensor data without the nuisance of a physically running motor, a browser-based visualizer was developed early in the process. This tool read incoming GSR values and rendered them in real time, and later allowed for tuning of the parameters with a more complex data scheme.

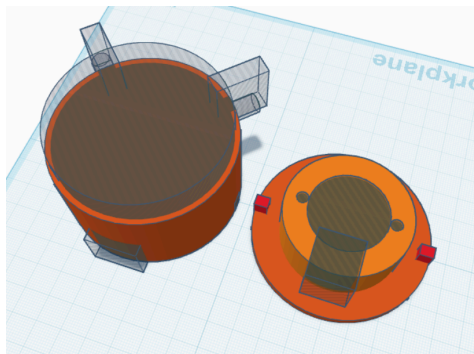


The EEG component provided by the Muse headset was integrated separately, then merged into the existing GSR scoring pipeline. The Muse outputs a processed anxiety score derived from its electrode readings, which could be tuned to be a primary, secondary, or equally weighted source along with the GSR. A host machine runs a Python script responsible for maintaining a Bluetooth connection to two Muse headsets and two Raspberry Pi Picos, all with corresponding hard-coded MAC addresses. The Pico handles GSR input and stepper motor output, and an HTTP API serves as the communication layer between the Python backend and the JavaScript visualizer running in the browser. Wiring this all together into a coherent data stream was a bit of a mess because of the language barrier between the Brainflow library and the MicroPython Bluetooth documentation. I choose to describe the process as educational.

Physical Prototyping

The physical prototyping phase focused on constructing the wig and integrating the mechanical system in a way that was both functional and wearable. While many of these elements were developed simultaneously, it is helpful to break them down into key components: the wig itself, the motor housing, the bang mechanism, the sleeve, and the glove. We also consistently kept the system as modular as possible, allowing for ongoing adjustments without needing to rebuild the entire structure. Each component could be tested independently without affecting the full system.

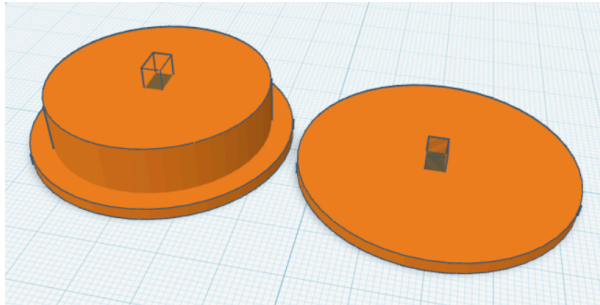
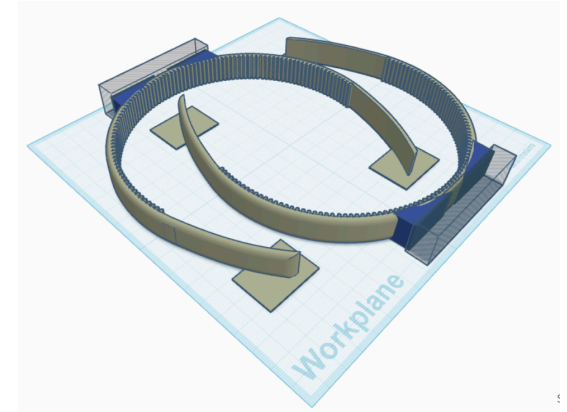
We began with the wig as a base object, using an inexpensive, pre-made, bright pink wig. We started with three very long wigs. The first one was used for experimentation and testing, and the other two were cut and reshaped to be shorter, in the interest of making them more manageable and to use the extra length for the creation of the bangs. This approach allowed us to experiment more freely with the bang mechanism without compromising the rest of the wig structure.



The motor housing and mounting system went through several iterations. Initially, the motor was intended to sit at the back of the head. However, we quickly realized that the weight of the motor is a factor that cannot be overlooked. Positioning it at the back created significant challenges in securing it in place, as the system would constantly be working against gravity. We shifted to placing the motor on top of the head, attached to a headband structure. The headband itself was sourced from an existing 3D model and slightly modified to stabilize the

motor holder, while the motor casing was designed by us. This positioning allowed for a more efficient transfer of motion and reduced strain on the system. In the final iteration, the motor casing was left open rather than enclosed with a lid, making it easier to access and adjust the string and bangs during testing.

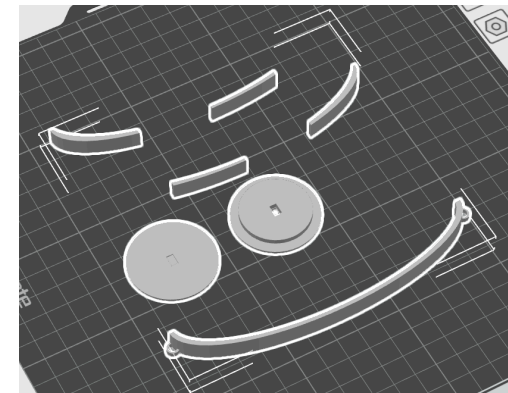
The spool mechanism attached directly to the motor was also developed through iteration. A simple 3D printed spool is used to wind and release the string. We went through a few iterations to adjust the hole size for the stepper motor shaft to ensure a snug fit and to increase the spool width and proportions for more stable reeling. It was also printed as two separate parts and then connected with a small amount of hot glue to ensure effortless printing.



The construction of the bangs themselves proved to be one of the more challenging aspects of the physical build. In order for the mechanism to function properly, the bangs needed to be lightweight and structurally sturdy enough. With the extra trial wig, a few different attempts were made. At first, we started with a thin, curved 3D printed band. We started with a mini version so that we could easily test different adhesives, like super glue, hot glue, PVA glue, and tape.

Through this process, we found that hot glue worked best due to its material properties. It was malleable in its in-between phase: when it was still warm enough to stick and be shifted around, but not so hot that it would burn our fingers as we adjusted and repositioned the hair slightly before it fully set, which was essential for achieving a consistent form.

However, we quickly ran into other issues. One major challenge was that the hot glue would dry relatively quickly, which meant we had to work more efficiently and be prepared to apply each tuft of hair



immediately once the glue was ready. In an attempt to manage this, we experimented with segmenting the band into four smaller sections so we could work in parts, but this felt awkward and was not structurally stable enough.

The biggest issue we encountered was finding the right balance of hair density. If the bangs were too thin, they lacked volume and structure. However, if they were thick enough to look visually complete, the hair would easily slip out or detach with minimal force, making the structure unreliable and not durable enough for its intended purpose. To solve this, we developed a layered construction method. Instead of applying thick sections of hair at once, we built the bangs up in thin layers: a layer of hot glue, a small amount of hair, another thin layer of hot glue, and then a strip of paper. The paper acted as a structural barrier, helping distribute the material without increasing the overall thickness of the band. This process was repeated to gradually build up the form. The opposite side was then secured using a similar continuous curved band that nested over the layered structure, enclosing it and keeping everything in place.

In addition to the main wig and mechanical components, we also developed two wearable support systems that were essential for the overall functionality of the artifact: a sleeve to hold the microcontroller and power supplies, and a glove to stabilize the GSR sensor placement.



We hand-sew the sleeve using a dark, semi-stretch knit fabric that we chose after examining a variety of different fabric types. It functions as a pouch worn on the participant's upper arm, secured with simple ribbon ties. Inside the sleeve, we housed the Raspberry Pi Pico, which handles the GSR sensor input and communicates with the stepper motor driver board, as well as two separate battery packs powering the system (one for the microcontroller and one for the motor).

We also constructed a component we refer to as the glove, similarly hand-sewn and attached using ribbon straps, which wraps around the participant's palm. This piece holds the

electrodes of the GSR sensor, with small openings in the fabric allowing the electrodes to maintain direct contact with the skin. Although GSR sensors are typically positioned on the fingers, we found through testing that placing the electrodes on the palm produced more consistent readings and, in some cases, more expressive variation in the data. As a result, the glove was designed to hold the GSR sensors on the participant's palm, while still allowing them to move their hand naturally during the experience.

Final Artifact & Observations

During the testing and final immersive experiences, we were able to observe different consistent patterns across participants. Overall, the system functioned as intended in terms of translating biofeedback into mechanical movement.

One of the most immediately clear observations was that the experience was generally perceived as playful and engaging. Participants often laughed and giggled or expressed curiosity when first interacting with the wigs, and there was a noticeable sense of ease and comfort that developed over time. The absurd nature of the artifact played an important and central role in this, because it reduced the initial tension between participants and to the artifact itself. It made the experience feel like a ridiculous shared experience between the pair.



The initial sense of awkwardness was transformed as a result of the combination of the extravagant wig, the visible mechanical movement, and the presence of another participant in the same situation. The original social discomfort quickly metamorphosed into a mutual recognition, a sort of camaraderie. The whimsical nature of the setup fostered a form of shared vulnerability, where both participants were equally exposed to the strangeness of the situation and to the other.

Interestingly, despite the visibility of the sensors and the system's output, the participant did not understand what caused the bangs to move either up or down. They did voice their curiosity and speculations about what the system was sensing, but they

did not puzzle over it for too long. They quickly accepted the mystery of the device they were wearing. They still engaged with it as an expressive or entertaining experience.



Once the system was explained, there was a noticeable behaviour change. Participants often began to actively try to “control” the bangs, either by calming themselves or by lightly stressing themselves out. They also tried to influence the other participant’s bangs by speaking to them, in order to either calm them or increase their stress levels. This created an unintentional but fascinating feedback loop; the system shifted from being a purely reactive object to something participants tried to control consciously.

Across sessions, it became clear that the artifact did not significantly alter participants’ core behaviour in terms of communication, but it did heighten their awareness of themselves within the interaction. The presence of visible biofeedback created a layer of self-consciousness, even when participants were not fully aware of the mechanics behind it. This aligns with the initial intention of making internal states externally perceptible, without fully prescribing how they should be interpreted.

Finally, while the system evolved significantly throughout the prototyping process, the final experience and intentions behind it remained consistent with the original conceptual purpose. The balance between playfulness and discomfort, vulnerability and disguise, endured, even as the technical implementation shifted and refined as the journey progressed.

Future Directions

While the current iteration of *InFringement* successfully demonstrates the translation of biofeedback into a shared performative experience, there remain several directions for future development. On a technical level, a deeper integration of physiological sensing could expand the range and nuance of the system's responsiveness. This version relies on a relatively surface-level interpretation of data from the GSR sensor and Muse headset. Honing how this data is processed or integrating more signals could allow for more subtle and varied forms of movement beyond the current up-and-down motion of the bangs.

Beyond the wig itself, the project could be extended into a broader system of wearable elements. Exploring other forms of gender expression through clothing, makeup, or bodily augmentation could create a more layered and immersive experience, where multiple aspects of presentation respond to the participant's internal state. This would allow the concept of gender performativity to be more clearly and deeply explored through a more complete embodied transformation.

There is also potential to further develop the social dimension of the work. Due to limitations, the current version only partially engages with the presence of an observing audience. Expanding the project into a more public or performative setting, where participants are knowingly watched and evaluated by a group of strangers, would most likely intensify the themes of judgment and self-awareness. This change would more directly engage with the relationship between private, hidden emotional states and visible vulnerability, which is at the core of this project.

Overall, future iterations would aim to deepen both the technical sensitivity and the social complexity of the experience, while maintaining the balance between playfulness and discomfort that defines the current artifact.

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